

1 Algorithm

Definition of Algorithm

An algorithm is a step-by-step set of instructions to solve a problem or perform a task.

Characteristics of Algorithm

- 1 **Finite** – must terminate after a limited number of steps.
- 2 **Definiteness** – each step is clear and unambiguous.
- 3 **Input** – takes zero or more inputs.
- 4 **Output** – produces at least one output.
- 5 **Effectiveness** – each step is simple and doable.

Examples of Algorithm

1. Find the sum of two numbers
2. Check whether a number is even or odd

Advantages of Algorithm

- Simple and easy to understand.
- Helps in problem solving.
- Language independent.
- Provides a clear step-by-step approach.

Disadvantages of Algorithm

- May be time-consuming for complex problems.
- Does not show the overall structure graphically.
- Can be difficult to maintain for large programs.

2 Flowchart

Definition of Flowchart

A flowchart is a graphical representation of an algorithm using standard symbols and arrows to show the flow of control.

Advantages of Flowchart

- Easy to understand and visualize.
- Helps in debugging and error detection.
- Improves communication.
- Useful for documentation.

Disadvantages of Flowchart

- Can become large and complex.
- Takes more time to draw.
- Not suitable for very detailed logic.
- Requires knowledge of standard symbols.

Examples of Flowchart

1. Flowchart to find the sum of two numbers
2. Flowchart to check whether a number is even or odd

3 Flowchart Symbols

Symbol	Name	Meaning
	Terminator (Start / End)	Shows the beginning or end of the flowchart.
	Process	Represents a calculation or action.
	Input / Output	Used for reading input or displaying output.
	Decision	Shows a condition (Yes / No).
	Flow Line / Arrow	Indicates the direction of flow.
	Connector	Used to connect different parts of a flowchart.

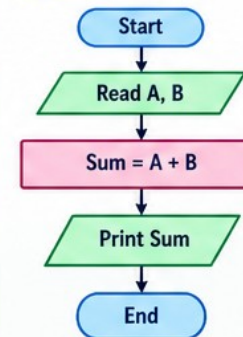
4 Examples

A. Find the Sum of Two Numbers

Algorithm

1. Start
2. Read first number (A)
3. Read second number (B)
4. Sum = A + B
5. Print Sum
6. End

Flowchart

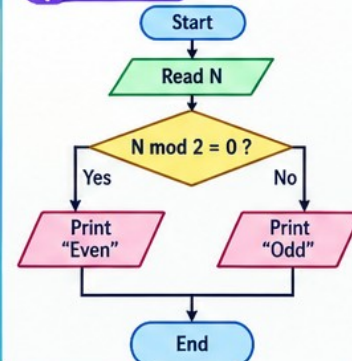


B. Check Whether a Number is Even or Odd

Algorithm

1. Start
2. Read number (N)
3. If $N \bmod 2 = 0$ then
 Print "Even"
 else
 Print "Odd"
4. End

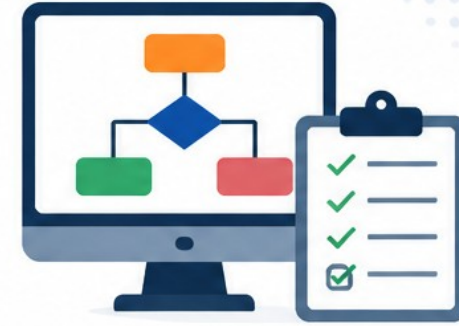
Flowchart



Algorithm



Algorithm is a **step-by-step procedure**, which defines a set of instructions to be executed in a certain order to get the desired output. Algorithms are generally created **independent of underlying languages**, i.e. an algorithm can be implemented in more than one programming language.



FROM THE DATA STRUCTURE POINT OF VIEW, FOLLOWING ARE SOME IMPORTANT CATEGORIES OF ALGORITHMS –



Search – Algorithm to search an item in a data structure.



Sort – Algorithm to sort items in a certain order.



Insert – Algorithm to insert item in a data structure.



Update – Algorithm to update an existing item in a data structure.



Delete – Algorithm to delete an existing item from a data structure.

CHARACTERISTICS OF AN ALGORITHM



An algorithm should have the following characteristics –



Unambiguous – Algorithm should be clear and unambiguous. Each of its steps (or phases), and their inputs/outputs should be clear and must lead to only one meaning.



Input – An algorithm should have 0 or more well-defined inputs.



Output – An algorithm should have 1 or more well-defined outputs, and should match the desired output.



Finiteness – Algorithms must terminate after a finite number of steps.



Feasibility – Should be feasible with the available resources.



Independent – An algorithm should have step-by-step directions, which should be independent of any programming code.

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ADVANTAGES OF ALGORITHM



It is a step-by-step repetition of a solution to a given problem which is very easy to understand.



Its easy to first develop an algorithm, then convert it into a flowchart & then into a computer program.



Its is easy to debug as every step is got its own logical sequence.

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DISADVANTAGES OF ALGORITHM



Writing algorithm takes a long time.



An Algorithm is not a computer program, it is rather a concept of how a program should be.

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EXAMPLE OF ALGORITHM



Let's try to learn algorithm-writing by using an example.



PROBLEM – Design an algorithm to add two numbers

Step 1 START

Step 2 declare three integers a, b & c

Step 3 define values of a & b

Step 4 add values of a & b

Step 5 store output of step 4 to c

Step 6 print c

Step 7 STOP



C PROGRAM

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int a, b, c;
6
7      // Step 3: define values of a & b
8      printf("Enter value of a: ");
9      scanf("%d", &a);
10
11     printf("Enter value of b: ");
12     scanf("%d", &b);
13
14     // Step 4: add values of a & b
15     c = a + b;
16
17     // Step 5: store output of step 4 to c
18     // Step 6: print c
19     printf("\nSum = %d", c);
20
21     return 0;
22 }
```



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CONT....



Algorithms tell the programmers how to code the program.
Alternatively, the algorithm can be written as –



ALGORITHM TO ADD TWO NUMBERS

Step 1 | START ADD

Step 2 | get values of a & b

Step 3 | $c \leftarrow a + b$

Step 4 | display c

Step 5 | STOP



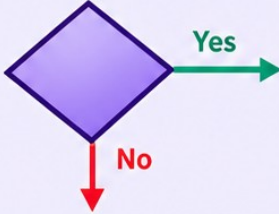
C PROGRAM (Using void main)

```
1  #include <stdio.h>
2  void main()
3  {
4      int a, b, c;
5
6      printf("Enter value of a: ");
7      scanf("%d", &a);
8      printf("Enter value of b: ");
9      scanf("%d", &b);
10
11     c = a + b;
12     printf("Sum = %d", c);
13
14     return;
15 }
16
```

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Flowchart **Symbols** and Their Purpose

Flowchart symbols	Geometric shape	Purpose
1 Ellipse		Ellipse is used to indicate the start and end of a flowchart. Start written in the ellipse indicates the beginning of a flowchart. End or Stop or Exit written in the ellipse indicates the end of the flowchart.
2 Parallelogram		A parallelogram is used to read data (input) or to print data (output).
3 Rectangle		A rectangle is used to show the processing that takes place in the flowchart.
4 Diamond		A diamond with two branches is used to show the decision making step in a flowchart. A question is specified in the diamond. The next step in the sequence is based on the answer to the question which is 'Yes' or 'No'.
5 Arrows		Arrows are used to connect the steps in a flowchart, to show the flow or sequence of the problem solving process.

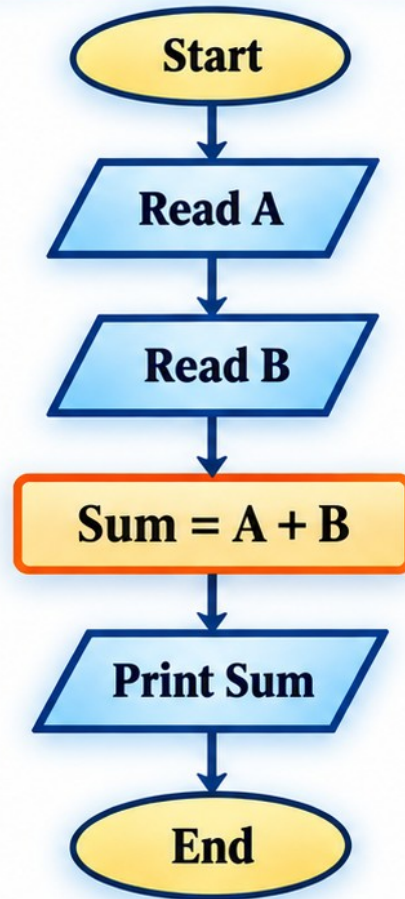


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Flowchart - How to find sum of two numbers

Flowchart



Finding sum of 845 and 247

- 1 Start
- 2 $A = 845$
- 3 $B = 247$
- 4 $\text{Sum} = 845 + 247$
- 5 $\text{Sum} = 1092$
- 6 End

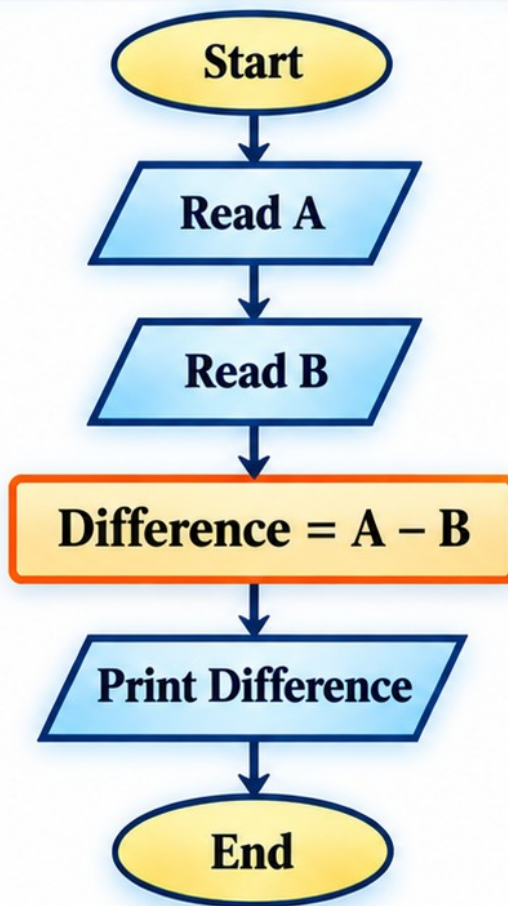


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Flowchart - How to find difference of two numbers

Flowchart



Finding difference of 845 and 247

- 1 Start
- 2 $A = 845$
- 3 $B = 247$
- 4 $\text{Difference} = 845 - 247$
- 5 $\text{Difference} = 598$
- 6 End



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ADVANTAGES OF USING FLOWCHARTS



- 1 Communication:**
Flowcharts are better way of communicating the logic of a system to all concerned or involved.



- 2 Effective analysis:**
With the help of flowchart, problem can be analyzed in more effective way therefore reducing cost and wastage of time.



- 3 Proper documentation:**
Program flowcharts serve as a good program documentation, which is needed for various purposes, making things more efficient.



- 4 Efficient coding:**
The flowcharts act as a guide or blueprint during the systems analysis and program development phase.



- 5 Proper Debugging:**
The flowchart helps in debugging process.



- 6 Efficient Program Maintenance:**
The maintenance of operating program becomes easy with the help of flowcharts, it help the programmer to put efforts more efficiently on that part.

DISADVANTAGES OF USING FLOWCHARTS



- 1 Complex logic:**
Sometimes, the program logic is quite complicated. In that case, flowchart becomes complex and clumsy. This will become a pain for the user, resulting in a waste of time and money trying to correct the problem.



- 2 Alterations and Modifications:**
If alterations are required the flowchart may require re-drawing completely. This will usually waste valuable time.



- 3 Reproduction:**
As the flowchart symbols cannot be typed, reproduction of flowchart becomes a problem.



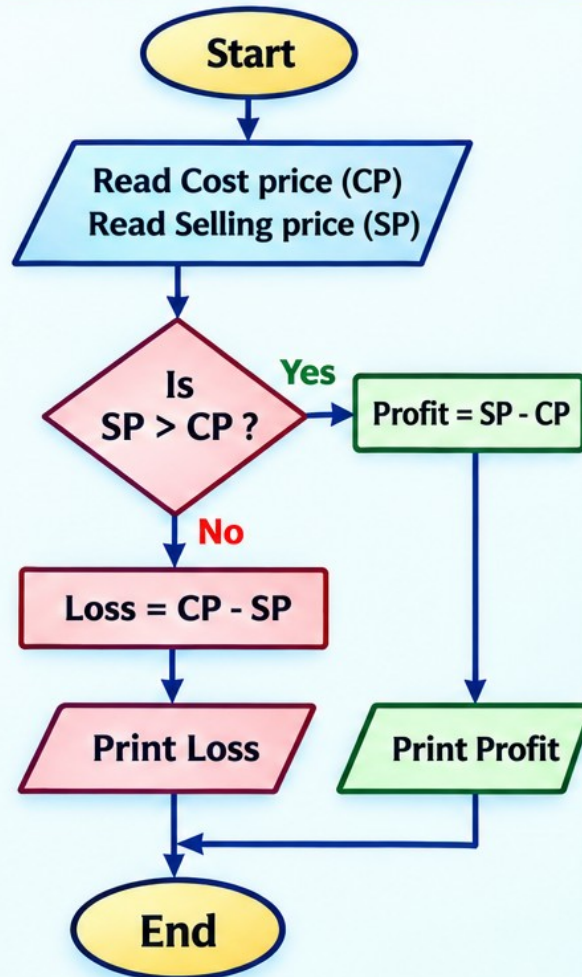
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Flowchart - How to find Profit or Loss

Simple Logic
Clear Solution

Flowchart



Algorithm / Pseudocode

- 1 Start
- 2 Read cost price (CP)
- 3 Read selling price (SP)
- 4 **If** SP > CP **then**
 Profit = SP - CP
 Print Profit
- 5 **Else**
 Loss = CP - SP
 Print Loss
- 6 End

Example

**Finding profit or loss when
CP = 325 and SP = 458**

- 1 Start
- 2 Read CP = 325
Read SP = 458
- 3 Check condition:
Is 458 > 325 ? — **Yes**
- 4 Profit = 458 - 325
= 133
- 5 Print Profit = Rs. 133
- 6 End

Output: Profit = Rs. 133

C Program (with int main and void main)

1. Using int main()

```
1 #include <stdio.h>
2 int main()
3 {
4     int cp, sp, profit, loss;
5     printf("Enter Cost Price (CP): ");
6     scanf("%d", &cp);
7     printf("Enter Selling Price (SP): ");
8     scanf("%d", &sp);
9     if (sp > cp) {
10         profit = sp - cp;
11         printf("Profit = Rs. %d\n", profit);
12     } else {
13         loss = cp - sp;
14         printf("Loss = Rs. %d\n", loss);
15     }
16     return 0;
17 }
```

2. Using void main()

```
1 #include <stdio.h>
2 void main()
3 {
4     int cp, sp, profit, loss;
5     printf("Enter Cost Price (CP): ");
6     scanf("%d", &cp);
7     printf("Enter Selling Price (SP): ");
8     scanf("%d", &sp);
9     if (sp > cp) {
10         profit = sp - cp;
11         printf("Profit = Rs. %d\n", profit);
12     } else {
13         loss = cp - sp;
14         printf("Loss = Rs. %d\n", loss);
15     }
16 }
```